Markov blankets and Bayesian mechanics (Karl Friston)

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welcome you are about to view a recorded talk of the mathematical consciousness science online seminar series held in 2020 this seminar series aims to explore the role of mathematics than the scientific study of consciousness and hopes to connect researchers who have an interest in this topic while every session of the seminar consists of a talk and discussions the latter are not recorded and the following will only contain the talk we hope you will enjoy it for further information please visit our website seminar math - consciousness org I'm going to go through the basic idea from or approach the basic ideas from a couple of perspectives just to put everything into context I want to in particular end up at this notion of a dual aspect information geometry and how that emerges and how it inherits from predictive processing or constructivist views of the brain and the second agenda is just to start to fine-grain or drill down on different kinds of inference that may map to different aspects of sentience and consciousness so that's where I'm trying to go let me start by a nod to where much where much of these ideas come from so I came into this really from the point of view of predictive coding and predictive processing as it's now currently known the idea of being here that we actively construct explanations for our sensorial so this is nicely illustrated by this oil painter in the 16th century famed for doing still lives that when viewed from another perspective have a very different interpretation so if before you saw a bowl of vegetables and now you see a face the point be made or the point that he was making is that this face this this hypothesis this fantasy for what cause these sensory impressions is coming from the inside out so this turns on its head the 20th century view of information processing where the information comes from the outside-in impresses itself on to use Helmholtz's was a nervous mechanism and somehow information is extracted to conclude there is a face and more looks at it from from an active or constructive point of view where you're checking or using the sensory information to check revise update your beliefs that you already have in mind about what could have caused those sensory inputs and indeed this is the exactly the view championed by Helmholtz objects are always imagined as being present in the field of vision as would have to be there in order to produce the same impression on the nervous mechanism that idea I think has had quiet traction since its introduction for example the ideas of Richard Gregory are about perception as iPods is testing the notion formerly has been taken up to great effect by people in machine learning like Geoffrey Hinton and Peter Dayan borrowing ideas from Richard Feynman specifically variational free-energy formulations and quantum electrodynamics and making that work or unpacking that in the context of the Bayesian brain or more generally an inference gloss on this ability to confirm your hypotheses about what caused your sensory data at this point in time so just taking that notion of sensory impressions what we have a hand is is the challenge of understanding how you and I make sense of the shadows on our sensory epithelia so what are causing those shadows that we are witness to impinging upon our eyes ears and indeed our interceptors the solution to that problem that inference problem you can just lift from standard Bayesian statistics or machine learning in the latter days and one ends up with a relatively simple scheme which can be construed in terms of something called Bayesian filtering in the neurosciences it's now become known as predictive coding but it is the same thing at all it's saying is that if I want to optimize my expectations the the belief I have about the causes of some sensory impression let's say that this was our sensory input and all I have to do is to minimize a quantity that we can write out as essentially a precision weighted prediction error so this equation here is a differential equation that dot means rate of change this means that this is an equation of motion or a flow so flow or like on a functional or free energy potential here there is a function of the expectation and the sensory impressions here and this is a form of a Kalman filter or predictive coding scheme so this is the prediction and these are the particular weighted prediction errors so what are the prediction the prediction errors well let's say we had this sensory input and we have this expectation about what could have caused this sensory impression here and the idea is that we can generate through a journey of model what we would have seen if this was correct and this provides a prediction that is subtracted from the sensation to form a prediction error such that if you were able to minimize the prediction error so that these two things were the saying you have effectively found an explanation or a hypothesis that is a sufficient explanation to explain your sensorial so the crucial point here that it is a sufficient explanation it's not necessarily the very little expression you will never know what's out there but if you can get through life minimizing your prediction errors then you've got a sufficiently good model to navigate and survive so that perspective provides a nice formal intuitive way of understanding perception it also enables you to accommodate action in the in this service of minimizing prediction error because if the imperative is say to minimize prediction error which will later become this free energy functional then there are two ways we can do that so if prediction errors sensations - predictions we can either change the predictions in our head to make them more like the sensations or we can change the sensations by acting upon the world to resample to palpate it in a different way to make the sensations more like the predictions so both action and perception by our changing sensations and predictions are just playing the same game and in hand trying to minimize that prediction error so we now have if like as an intuitive picture of active inference inference under action where you are in charge of harvesting those data that you predict so in some sense your actions now become self-fulfilling prophecies and we'll see an example of that later so that's the the other sort of the kind of introduction to these ideas I might give a psychology audience or a psychophysics audience and you're a biological audience what I want to do now is to change the tempo of the talk and just put that to one side and see if we can get to the same conclusions and the same kind of mathematical phenomenology from a purely first principle statistical approach and if we can what implications does that have for sentience so the way I'm going to do that is quickly go through a treatment of the statistics of light of self-organization with a special emphasis on Markov blankets and I've written a conic systems here and should really be non-equilibrium steady-state the implications of those non equilibrium steady states with Markov blankets in terms of information geometry and ago illustrate that with a numerical simulation of some active matter or a primordial soup and I'm going to close with different grades of sentience and active inference that I'm hoping will speak to different kinds of consciousness or the saw that you will be more familiar talking about so let's start with this question post by scrolling up how come the events in space and time which take place within the spatial boundary of a living organism be accounted for by physics and chemistry so we're not going to address that question head-on although we will implicitly answer it what we're going to do first of all is just focus on this notion of a spatial boundary and he would be the first person to a knowledge that boundary is itself a statistical object a construct and for those people who don't know what a Markov blanket is I'll just quickly take you through it imagine you've got some universe and these nodes of this graph here are states of some universe some systemic States and they are influencing each other or they have statistical or probablistic dependences denoted by the arrows here if I dis identify some set of states here McCullough's internal States then the Markov blanket comprises the parents the children and the parents and the children of these internal states and its role or its importance is that it provides a statistical shield or set of states that mean that if I wanted to predict what's going to happen to me my internal States now given the entire rest of the universe I will only ever need to know these blanket states that enshroud me here so all the information out there pertinent to what's going on in here is contained within the blanket states so technically these states are independent of these states given or conditioned upon the state of the blanket states now I'm going to make a further move which will make a lot more sense in the next slide and it's going to introduce a bipartition of the blanket States into active states that influence external states but do not influence internal states and sensory states that influence internal states that are not influenced by the internal states and you may be asking why have I done that well if one just looks at this probabilistic structure this very elemental statement of a dynamical world that somehow is coupled through dependencies and causal structures then we can look at these blanket States and in particular the partition into sensory active states in conjunction with the internal states that they encompass as the states of a particle and by a particle I mean something that could be as big as a grain or as small as an electron or something of the size of a cell all of these things particular things have in common a Markov blanket and in one sense they have two because if you didn't have this conditional independence you would not be able to measure in a statistical sense the difference between the thing or the particle and anything else it would become systole part of the universe so I'm just sketching out why that or how that unpacks for example in the brain we can consider all the internal brain States or the synaptic activity efficacy connectivity of inside the brain as the internal brain states and they influence active states or autonomous nervous system our our motor effectors that cause changes in the external states that then impinge back on the sensory states that in turn influence the internal states and the missing dependencies really here and here define this Markov blanket so there's an exchange between the internal and the outside but it has to be mediated with this certain with this bi-directional impermeable interface that is the Markov blanket where the sensory states mediate the inference of outside on the inside and the active States conversely the inside on the outside and exactly the same structure holds for a little cell where the surface states could constitute Li and the cell receptors could constitute the sensory states affecting the internal millia of the cell in terms of its intracellular processing and chemicals active States could be actin filaments that support the sensory states that can endow the cell with a motility that change the external Millia that again reports back in influences and sensory states so what we have here that inherits just from having a Markov blanket which every particle or thing must have is a circular causality that's very reminiscent of the action perception cycle I use the word sector causality deliberately because that's going to come back and do some heavy lifting later on right so what I want to do now is just put a bit of statistical physics on the Markov blanket but to do so I'm just I'm going to have to dis back off a little bit and ask you to forget about the Markov blanket for a moment and just bear with me why we why we well we drill down on self-organization in any arbitrary system then we're going to see what happens when we put the Markov blanket back in play again so what I've done here is just produce a little graphic of some universe or some system here just depicted in terms of two states one state here one state here so to any one point in time the state of the universe is a point and as time goes on it moves around in the state space tracing at a manifold or an attracting set that in fact is called methodically attracting set or a random dynamical attractor or a pull back attractor now the kinds of particles that were going to be talking about all systems are those that endure over time and in that instance that licensed me to understand the density of these trajectories as they unfold over time say for example I was getting up in the morning having my coffee going out to work well that's what I used to do before career for her Elizabeth but then you know following the my my normal trajectory over the day of course this could actually represent my heartbeat or it could be on a time scale over years we have Christmas and Eastern or some holidays irrespective of the time scale the nature of these states this kind of object must be in play and it means that I can interpret the density of these trajectories in terms of the probability that if you've sampled me at any one point in my heartbeat or any time during the day or during the year this also denotes a probability you'll find me in that state and I'm going to refer to that as a non-equilibrium steady state density now the important move that I can make just by stating or stipulative fining the kinds of systems we that we're interested in as dynamical systems that have this that these dynamics that conform to a long-run form and the Endura of a time means I can derive from this certain implications which actually take us back to that predictive coding scheme that we took we started off with so I won't go through this in detail but just to reassure you them there exists a simple mathematical description of the the way in which this non-equilibrium steady state density evolves over time as a function of the amplitude of random fluctuations on the flow F of the states as a function of where they are in state space and the peak corresponding to the this non-equilibrium steady state density here so this is a Fokker Planck equation you may know it as the master equation you may notice a scroller equation these are all equivalent formulations of this basic density dynamics ensemble dynamics for us we can do something quite useful here because we just said that these this object this probability density does not in and of itself change over time so we can set this to zero and we can look at the solution to this Fokker Planck equation and when we do so we see something very interesting what we see is that we can express the flow of states at any point in state space here as effectively a mixture of a gradient descent or in this integrated ascent on the log of this probability or over state space so this inherits from something called the Helmholtz decomposition this amplitude of the random fluctuations determines the degree to which I'm hill climbing this Q here denotes circuitous or Suren oil flow around probability gradients I'm gonna unpack that for you in the next slide but the key point to take on this is if things exist in particular those things with Markov blankets because to have a thing you have to have a Markov blanket then we can write down this kind of equation and this kind of equation essentially means you are doing a gradient flow or gradient ascent or descent on some quantity that has related the probability of me being in a particular state so for those people not familiar with the fokker-planck equation I just want to illustrate that imagine that I have a piece of ink and I'm dropping it into some solvents a glass of water and what I would expect to happen normally is that the ink molecules would dissipate due to the random fluctuations and distribute themselves over the entire volume of the solvent that's not the kind of system that we're interested in here though because we just said these systems gather themselves up together and are restricted to their attracting settler pull back attractor so this is what this kind of system would look like is it would look as if it was in down with some capacity to do inverse diffusion to actually diffuse up concentration gradients and that's exactly what this equation says what it's saying is that the flow in order for this attracting set to be realized to attain this non-equilibrium steady state has to have two components one it has to climb the gradients of the probability well strictly speaking the log probability of the kinds of states it is find found in and this Q term here is this circular flow solenoid flow around the I so probably contours basic structure the way to predicate the arguments on what I want to do with this slide is just convince you or at least provide some some graphical evidence that formalism that we've just briefly rehearsed underwrite nearly all of physics as we know it so special cases of this you will have done at school so when they amplitude of those random fluctuations on the flare goes to zero we arrive at conservative systems of the sort that would be useful for describing massive bodies gravitation Newtonian laws Lagrangian mechanics where it's all about this circular flow hence the planets revolving around the Sun and the moon around the earth we can look at the other special case where we eliminate the solenoidal flow or at least increase the amplitude of random fluctuations and then what we have is a mechanics that applies to fluctuating systems basically fluctuation dissipation theorem and all of statistical mechanics and statistical thermodynamics with the usual constants and proportionality here Boltzmann's constant and all the things you don't know about entropy and enthalpy again at school we can make another move we can just say somewhat arbitrarily although there are good reasons for doing it we can just express that non-equilibrium steady state density as the product of two complex roots and if we do that and following through we get quantum physics and this is actually just a restatement of that fokker-planck equation that we saw earlier on however if you remember the reason we're doing this is to see what are the implications of having a Markov blanket and this is where it starts to get more interesting from our perspective so there's another kind of physics out there there is unique to systems that have this by partition into internal and external separated by the blankets fades and that's as phasing mechanics which we've mentioned in terms of this perception action cycle so remember before that everything has to be described as this this solenoidal gradient flow on the negative on the logarithm of the the non equilibrium steady state density that has to be true for the internal states and the active States name in the blanket States which as I and the active States where the blanket States comprises sensory in the active States notice what I've also done here I've swapped out the log probability here for a free energy functional that is a function of the blanket States and this object here which will become quite crucial later on which is a a probability distribution over the internal states that is parameterized by the expected internal States a bit subtle though but that's going to come very important later on but the two key things that emerge from this relatively straightforward analysis first of all the internal and active states which we'll call autonomous States share the same function so it'll look as if my internal brain and the way I act and try to do the same thing so we come back this notion that they're both trying to minimize say prediction error or in this instance more generally minimizing by this gradient flow this variational free energy functional clearly because of the structure of the Markov blanket this variation free energy does not depend upon the external States so notice that this thing here is not a function of the external state is a function of a function of the external states that parameterize by the internal states so I'll just preempt or foreshadow some of the due respect geometry conclusion so what we have here is a gradient flow that can be written down as a functional not of external States but as a function of a stern states but as a functional of beliefs about probabilistic beliefs about external states that is never in and of itself directly influenced by the external states but just what the internal space in that think about or parameterize what the internal States parameterize in terms of a belief distribution over the external states so that's a bit heavy so let me make it a little bit lighter so this free energy functional could have many interpretations and clearly on the present analysis undergirds all of the self-organization that we've been talking about I'm just making the point here I can now try to go through this quickly the point I'm it is being made here is that this basic structure this basic mechanics were internal states of a system and their action upon their world can be construed or it looks as if they are minimizing this free energy functional here of the blanket date in their sensory data and what they currently encode it looks very much like for example action and beliefs trying to minimize free energy or maximize the negative of that basically value where value is simply the probability of a low probability of your blanket States being in a particular configuration so this if you like can be viewed as a sort of a mathematical backdrop to reinforcement learning optimal control theory in economists expected utility theory if on the other hand you had been brought up in terms of information theory then this value function the law of probability the kinds of blanket states I unlikely to be found in the ones I prefer to be found in that define me understand the negative of that logarithm that's known as self information and information theory treiber's call it surprisal we'll just call it surprised this free energy quantity is another bound on that surprise here and if we think of this optimized aid from the point of view of optimization what we are saying is that we're trying to minimize our surprise and in so doing we can unpack that in terms of things like integration information theory minimum redundancy principle of Horace Barlow and indeed the free energy principle that we're going to casually mention throughout this presentation that's nice because the expected value if I do this on average over time the expected value of self information or surprises entropy which means it looks as if this minimization take on this gradient flow will be in the service of resisting the dissipation and decay and disorder entailed by having a high entropy and it'll look as if it's consistent with the principles of self-organization or synergetics if you're in laser physics or if you're physiologist they look by homeostasis just keeping things within bounds minimizing homeostatic surprises if you're a statistician you would look at this conscious AR this is a probability of some blanket States or data that I have access to from the app that are a product of the outside world under some model here so hitherto this has been a Markov blanket but now I'm gonna say no way this is actually we know from a statistical point of view it's going to stand in for a model and this is known as Bayesian model evidence also known as the marginal likelihood because I've marginalized out all the external causes of my blanket States here so then we can sort of speak to people in the like the brazing Bayesian brain hypothesis people who look at the brain as accumulating evidence and get back to predictive processing and it's special cousins or variants such as predictive coding oh yeah so the Markov blanket this was sent to me by one of my young colleagues who moved over to America he's just had a little baby and his wife bought him a Markov blanket so if you want to Markov blanket you can get these in America so here's a little Kyra making our inferences first inferences about the world through through her own mark that's Markov on the blanket itself right so information geometry I'm gonna use numerical simulations here just to show how these things arise and how you can think about what we've talked about in terms of that a Bayesian mechanics in terms of something called generalized synchrony so but let me just take you through this with it with an illustration what we've done here is just create a little universe a primordial soup by writing down equations of motion for 128 little particles or macromolecules each of which has their autonomous dynamics in fact based upon a Lorentz attractor and we've equipped these this ensemble of synthetic molecules with strong and weak attractions and repulsions the details are irrelevant you can generally get this kind of behavior from any set of differential equations the reason we wrote these down is that now we know exactly the influences that one particle exerts over another particle and because we know that we can actually go in and identify a Markov blanket and then ask the question is there a little Markov blanket or a little thing or a particle inside this soup and if so does it look as if it's conforming to the Bayesian brain or conforming to houris borrows principle of minimum redundancy so I'm showing the same data here but where I've color-coded all of those molecules once they have attained non-equilibrium steady state as to whether there are external states that are hidden beyond the Markov blanket so therefore sometimes referred to as hidden States sensory states that as is typical in these simulations are under written or above the active states in red then cells enshroud the internal states what we see here is something that's not unlike a little cryin or viral particles their particle just wriggling around in its primordial soup having attained this non-equilibrium steady state and coursing in terms of its state space over this attracting set so this allows us to ask a question do the internal states the blue states appear to infer the causes of their sensory states the magenta states in other words are they do they look as if they're doing a gradient flow on this model evidence as Bayesian model evidence sometimes referred to by people like a top who are self evidencing and indeed they do it's illustrated in this graphic here so what we've done here is find a mixture of internal states that predicts a mixture of the motion of the external states so this will be a little bit like finding a neuronal population or a mixture of neuro neurons in a small brain that responded to say motion in the outside world communicated by via the sensory state so the visual epithelium and when we look for this we try to identify those mixtures that maximally show a correlation we do indeed find that the internal states are in some way parameterizing or encoding or representing the behavior of the external states despite the fact they can't see them directly so what we've done here is just plot the prediction of the external state as a function of the internal state the expected internal state over time along with ninety percent confidence intervals and the actual external motion in cyan here and you can see that there's a pretty good inference going on and interestingly remember before I was talking about the expected internal states that capital mu that actually is important because of what it allows me to do is to sort of look at the expected internal states and expected external states and just see this sync as synchronization manifold that on average when the internal states are like this the external states are roughly like this if I go through and is pick out excursions here and overlay themselves on each other to get at these averages here what we see is something very much like what we see in electromagnetic as that is of the human brain and cognitive neuroscience for example which is a simulated event-related potentials there's a one from the empirical literature and this is that is overlaying these fluctuations of internal states that are themselves predictive of external states so I've used the term synchronization manifold here deliberately because we're all we're looking at here is something that has been known for centuries in fact introduced the notion wasn't used by history no Christian equivalents as synchronization of chaos and generalized synchronization so for those of you who haven't come across this before it's most famously articulated by the observation that when you have many clocks suspended from the same wall or the same beam in terms of engineering we call them loosely coupled dynamical systems they will never tably come to swing in unison and in synchrony so that is the only attracting set that they can possibly occupy and that exactly describes this generalized synchrony between the outside and the inside that we're witnessing here so from our point of view this is actually a drawing by Huygens himself of two clocks suspended from the same beam one clock we think all the internal states the outside world can be the other clock and the beam become the blanket states with the active and sensory components here I like this perspective because it highlights the pure symmetry of this structure so if you subscribe to the notion that you're trying to infer your universe then you also are committed to the idea that your universe is actually trying to infer you which actually is an interesting perspective when it comes to things like eco niche construction like anyway just looking at the that synchronization manifold a little bit more closely it has within it a structure which holds the key to this dual aspect information geometry that I have repeatedly mentioned and it's a very simple consequence if I simply plot a little state space with internal states here in external states there and I do so for every given blanket state say sensory input then it has to be the case there must be a probability distribution for any given blanket State over the internal states and the external states now it's self-evident it's obvious but it has a fundamental implication what it means is that for every blanket state there is an expected internal state that effectively via this synchronization manifold Karamat rises a belief distribution over the external states so just by having a Markov blanket I'm conditioning the inside and the outside on any given blanket state it must be the case that on average the internal states in some sense represent the external states and we can formalize that in terms of information geometry I just put this slide up if you want to talk about what is information geometry we can do that this slides like this panel is just meant to intimate that there are there are two kinds of information geometry is going on here there's a kind of information geometry that you would be dealing with if you were a physicist dealing with the statistical thermodynamics of this little soup it's a probability distributions over the internal states in and of themselves and from that you can spin off dissipation fluctuation dissipation integral fluctuation theorems the second law of thermodynamics should you want to and there's another kind of information geometry going on here because for every expected internal state there's another probability distribution over external States about external States that has its own information geometry which means as I move around in my internal state space I am moving in two spaces at the same time one is the space of the Laika configurations on average that I will occupy on the inside but at the same time I'm moving around in an information geometry which is about things on the outside the causes of my grandkids States or my sensorial so in summary internal space parameterize probabilistic beliefs about external States and equip them with a dual aspect geometry so let me just summarize we've done all the heavy lifting now in this talk and so let's summarize where we come from as you know under this physicists take on self-organization of things that are equipped with Markov blanket the existence of a Markov blanket necessarily implies a partition of states into internal states and their blanket States no miss sensor United States and external states that are hidden behind the blanket and because active States change but are not changed by external states they will appear to minimize the entropy of external states and their Markov blanket and this means that action will appear to minimize the structural and functional integrity of the Markov blanket and this has been framed very elegantly by people like varela and his colleagues in terms of auto piecess internal states appear to infer the hidden causes of sensory states by maximizing amazing model evidence and influence those causes through action and we would put a bathing gloss on that in terms of an inactive kind of inference and in virtue of the implicit generalized synchrony there must be there if there's a Markov blanket there must be there if something exists there exists a dual aspect information geometry in which expected internal states parameterised conditional beliefs / external states again conditioned upon the Markov blanket so let me just close now by taking the considerations to the next level predicated on this representational or dual aspect geometry and see what they might mean for sentience but to do that I've got to if you like provide a taxonomy of different kinds of self evidencing that you get from this framework so here's a Markov blanket again just to remind you we've got our Markov blanket partition here with the mark of blankets and we can lump the the internal states and the active states as these sort of free energy minimizing prediction error minimizing states that lend a system a degree of autonomy and call them autonomous states under by alpha in a minute the blanket states comprise the sensor net if states and then together these can be thought of as the particle in question either a brain or a the sinners don't look at the equations what I want to do here is just show you where two different ways of unpacking this phasing mechanics come from depending upon how far you want to go in terms of the mathematical formalism so largely what we've done so far is start off with the assumption that there is a universe that has Noren dynamics with random fluctuations and flows and we've quantified or enumerated the amplitude of these random fluctuations in terms of this gamma here this inverse precision and if this is true at non equivalent steady state when things have characteristics from measured over time then the solution requires this relationship to be true the flow is related to the gradients of surprise if the thing in addition has a Markov blanket I'm stupid to believe so it must have then this applies to this and we had now have this hand-in-hand internal perception like dynamic an active dynamic both in the surface of minimizing or minimizing surprisal that we can write in terms of this variational function that depends upon this parameterised belief about the external states mathematically that's a a variational density or approximate posterior density parameterize by the internal states and we can write down things like the free energy principle as a principle a variational principle of least action where the action is in this instance denoted by the active States but action also has an another meaning action in physics means the time or the path integral of energy so there's a whole other layer to this kind of argument and which I'm going to show you in a second but let me just see how far we can get with this kind of formalism so if I commit to that those dynamics and just write down some of these probability distributions and just integrate those equations of motion then you can get quite a long way in terms of simulating active sentence or active inference so what we've done here is is just write down a generative model in terms of that probability over external states and the blanket the consequences the blanket States and so then we can derive a free energy functional write down the equations of motion in this instance just for your interest this little agent has the belief that there is an invisible object point that's pulling its finger around a trajectory that is itself prescribed by a neurons attractor and by suitably configuring some nonlinear mappings when it acts to fulfill those beliefs to minimize his prediction errors in this instance the predictions about what he should see and feel that it's actually doing so it's causing its sensations by moving its hand around we can get it to simulate things like writing and because we are generating that writing on that action we can think of this in terms of action but also action observations so by replaying this in the absence of any proprietary predictions we can start to simulate action observation and mirror neuron like behaviors in silico I won't go into any details there or I'm wanted to do with this slide is say that just with these very fast gradient flows of a predictive processing predicted coding like flavor you can get quite a long way in terms of simulating autonomous behavior both in terms of inferring and acting and indeed inferring what you see being enacted but there's a different kind of autonomous behavior that may be important for understanding the way that we actually operate as opposed to say an insect or a virus or a thermistor might might operate and that's in terms of action as a pathogen go into the future so the whole other bunch of physics you can bring to the table here again leveraging the contributions of richard fineman in terms of the path integral formulation of quantum electrodynamics we can tell this same story but not in terms directly of the fokker-planck equation but it's equivalent in terms of a path integral formulation where now we're talking not about variational free energies but variational actions that are integrals of this free energy functional or it's its basis which is the surprisal over time and this can be decomposed into all sorts of interesting components that kinetic energy have an independent path dependent terms that's from our point of view not so interesting it's the homologue the ensues when we adopt the Markov blanket and what comes out of that now is a slightly different perspective on the way that we can look as if we are minimizing the path integral of variational free energy and this leads us to the notion of planning as inference it leads us to choosing actions now that we think will in the future have a trajectory or path through state space that is the least surprising on average or conforms to our prior beliefs about the sorts of states that I will encounter in the future so now but simply we have a statistical physics explanation or calculus for planning and this is the last one last slides no this is just an illustration when you put that kind of mechanics in silico and simulate action now not simply doing the gradient descent on variational free energy but actually choosing paths of action into the future planning to minimize on average they expected variation for energy in the future you can start to simulate things like saccadic searches hunting for information it's going to reduce the expected surprise uncertainty by looking over there as opposed to looking over there in brief this simulation is a showing the the the evidence accumulation through an epistemic foraging of the visual field that is mediated by little saccadic eye movements gathering information to resolve uncertainty about these conditional beliefs about the causes the external states of the sensory input here disambiguating between upright sideways and an inverted face so this slide summarizes the if you like the end point of this most elaborated form of free energy minimization what it says is you can think of this as so senses sensory information coming in you evaluate your or you change your internal states to optimize your beliefs about external states through this variational or approximate posterior conditional belief distribution here simply by minimizing free energy that can be broken down into in accuracy in complexity and then you use that to work out what would happen in the future if I did this all that and that entails a minimization or that selection entails a minimization of the expected free energy and interesting the expected inaccuracy becomes ambiguity the expected complexity becomes risk so now what we have is a calculus from first principles that gives you a description of agents that are risk minimizing and information seeking in the sense that they want to minimize their ambiguity and indulge in that kind of visual palpation that does this epistemic foraging or responding to epistemic affordances and we saw in the previous slide so in summary well this is not a summary this is levels of conversation that we can now have hopefully if we have time about how far you can take this and what different levels of garrotted models would be required to support different kinds of active engagement with your world via sensations that could be assets or ium could have sentience and what does what was the difference between an unconscious inference in the sense of hellhole and literary thinking and planning about the future does that require deep genitive models and the consequences of action what does that mean for autonomy and agency how do things become how do we lose the normal transparency we haven't talked about a precision but that may be an important aspect this encoding uncertainty in these belief distributions of external states and what does it mean to have the hypothesis the fantasy that I am myself and does this underwrite subjective consciousness and self-awareness so these are questions that I don't have answers to but I thought might be the kinds of questions that at least some people in the audience might want to might want to speak to and so I'll give the last word to helmoltz again objects always imagined as being present in the field of vision as would have to be there in order to produce the same impression on the nervous mechanism so with that I just wanted to thank those people whose ideas that I've been talking about and of course thank you for your attention thank you very much indeed you